



## SAFETY DATA SHEET

### KAY-5 SANITIZER

MSDS No.:

AA00280-0915051020

#### 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : 케이-5 세니타이저  
KAY-5 SANITIZER

Other means of identification : Not applicable.

Recommended use : Sanitizer - Food contact surface

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.3 %

Company : ECOLAB KOREA LTD.  
8F, Daemyung Tower  
135 Beobwon-ro, Songpa-gu, Seoul Korea 05836  
82-1588-7595

Emergency telephone number : 82-1588-7595

Issuing date : 16.02.2021

#### 2. HAZARDS IDENTIFICATION

##### GHS Classification

##### Product AS SOLD

Acute toxicity (Oral) : Category 4  
Serious eye damage/eye irritation : Category 2  
Chronic aquatic toxicity : Category 2

##### Product AT USE DILUTION

Not applicable.

##### GHS Label element

##### Product AS SOLD

Hazard pictograms :



Signal Word : Warning

Hazard Statements : H302 Harmful if swallowed.  
H319 Causes serious eye irritation.  
H411 Toxic to aquatic life with long lasting effects.

Precautionary Statements : **Prevention:**  
P264 Wash the contact area thoroughly after handling.  
P270 Do not eat, drink or smoke when using this product.  
P273 Avoid release to the environment.  
P280e Wear eye protection/face protection.  
**Response:**

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P301 + P312 + P330 IF SWALLOWED: Call a POISON CENTER or doctor/ physician if you feel unwell. Rinse mouth.

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.

P337 + P313 If eye irritation persists: Get medical advice/ attention.

P391 Collect spillage.

#### **Disposal:**

P501 Dispose of contents and container according to wastes control act.

#### **Product AT USE DILUTION**

Hazard pictograms : Not applicable.

Signal Word : Not applicable.

Hazard Statements : Not applicable.

Precautionary Statements : **Prevention:**  
P264 Wash hands thoroughly after handling.  
**Response:**  
P314 Get medical advice/ attention if you feel unwell.  
**Storage:**  
P401 Store in accordance with local regulations.

#### **Product AS SOLD**

Other hazards : Mixing this product with acid or ammonia releases chlorine gas.

### 3. COMPOSITION/INFORMATION ON INGREDIENTS

#### **Product AS SOLD**

Pure substance/mixture : Mixture

| Chemical Name                               | Common name       | CAS-No.    | Concentration: (%) |
|---------------------------------------------|-------------------|------------|--------------------|
| Sodium Carbonate                            | no data available | 497-19-8   | 10 - 20            |
| Sodium dichloro-s-triazinetri-<br>dihydrate | no data available | 51580-86-0 | 1 - 10             |
| glycerin                                    | no data available | 56-81-5    | 0.1 - 1            |

#### **Product AT USE DILUTION**

No hazardous ingredients

All unlisted components do not contribute to the overall hazard of the product and are trade secret, making up 100% of the full composition.

### 4. FIRST AID MEASURES

#### **Product AS SOLD**

In case of eye contact : IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. If eye irritation persists: Get medical advice/ attention.

In case of skin contact : Rinse with plenty of water.

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|                                                             |                                                                                                   |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| If swallowed                                                | : Rinse mouth. Get medical attention if symptoms occur.                                           |
| If inhaled                                                  | : Treat symptomatically. Get medical attention if symptoms occur.                                 |
| Protection of first-aiders                                  | : If potential for exposure exists refer to Section 8 for specific personal protective equipment. |
| Notes to physician                                          | : Treat symptomatically.                                                                          |
| Most important symptoms and effects, both acute and delayed | : See Section 11 for more detailed information on health effects and symptoms.                    |

#### Product AT USE DILUTION

|                         |                                                         |
|-------------------------|---------------------------------------------------------|
| In case of eye contact  | : Rinse with plenty of water.                           |
| In case of skin contact | : Rinse with plenty of water.                           |
| If swallowed            | : Rinse mouth. Get medical attention if symptoms occur. |
| If inhaled              | : Get medical attention if symptoms occur.              |

### 5. FIREFIGHTING MEASURES

#### Product AS SOLD

|                                               |                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Suitable extinguishing media                  | : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.                                                                                                                                                                          |
| Unsuitable extinguishing media                | : None known.                                                                                                                                                                                                                                                                      |
| Specific hazards during firefighting          | : Not flammable or combustible.                                                                                                                                                                                                                                                    |
| Hazardous combustion products                 | : Decomposition products may include the following materials:<br>Carbon oxides<br>Sulphur oxides                                                                                                                                                                                   |
| Special protective equipment for firefighters | : Use personal protective equipment.                                                                                                                                                                                                                                               |
| Specific extinguishing methods                | : Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes. |

### 6. ACCIDENTAL RELEASE MEASURES

#### Product AS SOLD

|                                                                     |                                                                                                                    |
|---------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Personal precautions, protective equipment and emergency procedures | : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8. |
| Environmental precautions                                           | : Do not allow contact with soil, surface or ground water.                                                         |
| Methods and materials for containment and cleaning up               | : Sweep up and shovel into suitable containers for disposal.                                                       |

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#### Product AT USE DILUTION

- Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : No special environmental precautions required.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

### 7. HANDLING AND STORAGE

#### Product AS SOLD

- Advice on safe handling : Do not ingest. Avoid contact with skin and eyes. Do not breathe dust/fume/gas/mist/vapours/spray. Wash hands thoroughly after handling. Mixing this product with acid or ammonia releases chlorine gas. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).
- Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.
- Storage temperature : 0 °C to 40 °C

#### Product AT USE DILUTION

- Advice on safe handling : In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).
- Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

### 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Product AS SOLD

##### Components with workplace control parameters

| Components | CAS-No. | Form of exposure | Permissible concentration | Basis  |
|------------|---------|------------------|---------------------------|--------|
| glycerin   | 56-81-5 | TWA (Mist)       | 10 mg/m <sup>3</sup>      | KR OEL |

- Biological occupational exposure limits : no data available
- Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

#### Personal protective equipment

- Eye protection : Safety glasses with side-shields
- Hand protection : No special protective equipment required.

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|                        |                                                                                                                                           |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Skin protection        | : No special protective equipment required.                                                                                               |
| Respiratory protection | : No personal respiratory protective equipment normally required.                                                                         |
| Hygiene measures       | : Handle in accordance with good industrial hygiene and safety practice. Wash face, hands and any exposed skin thoroughly after handling. |

#### Product AT USE DILUTION

|                      |                                                                                                      |
|----------------------|------------------------------------------------------------------------------------------------------|
| Engineering measures | : Good general ventilation should be sufficient to control worker exposure to airborne contaminants. |
|----------------------|------------------------------------------------------------------------------------------------------|

#### Personal protective equipment

|                        |                                                                   |
|------------------------|-------------------------------------------------------------------|
| Eye protection         | : No special protective equipment required.                       |
| Hand protection        | : No special protective equipment required.                       |
| Skin protection        | : No special protective equipment required.                       |
| Respiratory protection | : No personal respiratory protective equipment normally required. |

## 9. PHYSICAL AND CHEMICAL PROPERTIES

|                                         | Product AS SOLD                                 | Product AT USE DILUTION |
|-----------------------------------------|-------------------------------------------------|-------------------------|
| Appearance                              | : powder                                        | liquid                  |
| Colour                                  | : opaque, white                                 | colourless              |
| Odour                                   | : Chlorine                                      | Chlorine                |
| pH                                      | : 10.0 - 10.4, (1 %)                            | 10.5                    |
| Flash point                             | : Not applicable., Does not sustain combustion. |                         |
| Odour Threshold                         | : no data available                             |                         |
| Melting point/freezing point            | : no data available                             |                         |
| Initial boiling point and boiling range | : > 100 °C                                      |                         |
| Evaporation rate                        | : no data available                             |                         |
| Flammability (solid, gas)               | : Not applicable.                               |                         |
| Upper explosion limit                   | : no data available                             |                         |
| Lower explosion limit                   | : no data available                             |                         |
| Vapour pressure                         | : no data available                             |                         |
| Relative vapour density                 | : no data available                             |                         |
| Relative density                        | : 1.0 - 1.5                                     |                         |
| Water solubility                        | : soluble                                       |                         |
| Solubility in other solvents            | : no data available                             |                         |
| Partition coefficient: n-octanol/water  | : no data available                             |                         |
| Auto-ignition temperature               | : no data available                             |                         |
| Thermal decomposition                   | : no data available                             |                         |

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|                      |                                                            |
|----------------------|------------------------------------------------------------|
| Viscosity, kinematic | : no data available                                        |
| Explosive properties | : no data available                                        |
| Oxidizing properties | : The substance or mixture is not classified as oxidizing. |
| Molecular weight     | : no data available                                        |
| VOC                  | : no data available                                        |

### 10. STABILITY AND REACTIVITY

#### Product AS SOLD

|                                    |                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------|
| Reactivity                         | : No dangerous reaction known under conditions of normal use.                                    |
| Chemical stability                 | : Stable under normal conditions.                                                                |
| Possibility of hazardous reactions | : Mixing this product with acid or ammonia releases chlorine gas.                                |
| Conditions to avoid                | : None known.                                                                                    |
| Incompatible materials             | : Acids<br>Metals                                                                                |
| Hazardous decomposition products   | : Decomposition products may include the following materials:<br>Carbon oxides<br>Sulphur oxides |

### 11. TOXICOLOGICAL INFORMATION

|                                          |                                         |
|------------------------------------------|-----------------------------------------|
| Information on likely routes of exposure | : Inhalation, Eye contact, Skin contact |
|------------------------------------------|-----------------------------------------|

#### Potential Health Effects

#### Product AS SOLD

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Causes serious eye irritation.                              |
| Skin             | : Health injuries are not known or expected under normal use. |
| Ingestion        | : Harmful if swallowed.                                       |
| Inhalation       | : Health injuries are not known or expected under normal use. |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

#### Product AT USE DILUTION

|                  |                                                               |
|------------------|---------------------------------------------------------------|
| Eyes             | : Health injuries are not known or expected under normal use. |
| Skin             | : Health injuries are not known or expected under normal use. |
| Ingestion        | : Health injuries are not known or expected under normal use. |
| Inhalation       | : Health injuries are not known or expected under normal use. |
| Chronic Exposure | : Health injuries are not known or expected under normal use. |

#### Experience with human exposure

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#### Product AS SOLD

|              |                                  |
|--------------|----------------------------------|
| Eye contact  | : Redness, Pain, Irritation      |
| Skin contact | : No symptoms known or expected. |
| Ingestion    | : No symptoms known or expected. |
| Inhalation   | : No symptoms known or expected. |

#### Product AT USE DILUTION

|              |                                  |
|--------------|----------------------------------|
| Eye contact  | : No symptoms known or expected. |
| Skin contact | : No symptoms known or expected. |
| Ingestion    | : No symptoms known or expected. |
| Inhalation   | : No symptoms known or expected. |

#### Toxicity

##### Product AS SOLD

##### Product

|                                   |                                                                          |
|-----------------------------------|--------------------------------------------------------------------------|
| Acute oral toxicity               | : LD50 rat: $\geq$ 2,000 mg/kg                                           |
| Acute inhalation toxicity         | : 4 h Acute toxicity estimate : $>$ 5 mg/l<br>Test atmosphere: dust/mist |
| Acute dermal toxicity             | : no data available                                                      |
| Skin corrosion/irritation         | : no data available                                                      |
| Serious eye damage/eye irritation | : no data available                                                      |
| Respiratory or skin sensitization | : no data available                                                      |
| Carcinogenicity                   | : no data available                                                      |
| Reproductive effects              | : no data available                                                      |
| Germ cell mutagenicity            | : no data available                                                      |
| Teratogenicity                    | : no data available                                                      |
| STOT - single exposure            | : no data available                                                      |
| STOT - repeated exposure          | : no data available                                                      |
| Aspiration toxicity               | : no data available                                                      |

##### Components

|                       |                                                                           |
|-----------------------|---------------------------------------------------------------------------|
| Acute dermal toxicity | : Sodium dichloro-s-triazinetriene dihydrate<br>LD50 rat: $>$ 5,000 mg/kg |
|                       | glycerin<br>LD50 rabbit: 23,000 mg/kg                                     |

## 12. ECOLOGICAL INFORMATION

##### Product AS SOLD

##### Toxicity

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Environmental Effects : Toxic to aquatic life. Toxic to aquatic life with long lasting effects.

#### Product

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

#### Components

Toxicity to fish : Sodium Carbonate  
96 h LC50 Lepomis macrochirus (Bluegill sunfish): 300 mg/l  
  
glycerin  
96 h LC50 Fish: 855 mg/l

#### Components

Toxicity to daphnia and other aquatic invertebrates : Sodium Carbonate  
48 h EC50 Ceriodaphnia (water flea): 213.5 mg/l  
  
Sodium dichloro-s-triazinetriene dihydrate  
48 h EC50 Daphnia: 0.196 mg/l

#### Persistence and degradability

##### Product AS SOLD

Readily biodegradable.

##### Product AT USE DILUTION

Readily biodegradable.

#### Bioaccumulative potential

no data available

#### Mobility in soil

no data available

#### Other adverse effects

Ozone layer hazard: not applicable

### 13. DISPOSAL CONSIDERATIONS

##### Product AS SOLD

Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

##### Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.



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Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

#### 14. TRANSPORT INFORMATION

##### Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

##### Land transport (KR DG)

Not dangerous goods

##### Sea transport (IMDG/IMO)

Not dangerous goods

Special precautions for user : None

#### 15. REGULATORY INFORMATION

##### National regulatory information

##### Industrial Safety and Health Act

##### Harmful Substances Prohibited from Manufacturing

Not applicable.

##### Korea. Harmful Substances Requiring Permission

Not applicable.

##### Substances to be controlled (Article 442 of the Regulation on Industrial Health Standards)

Not applicable.

##### Controlled Substances Subject to Environment Monitoring

Not applicable.

##### Controlled Substances Subject to Health Examination

Not applicable.

##### Substances with Workplace Exposure Limits

glycerin

##### Chemicals Control Act

##### Toxic chemicals

Not applicable.

##### Authorized chemicals

Not applicable.

##### Restricted Toxic Chemicals

Not applicable.

##### Banned Toxic Chemicals

Not applicable.

##### Toxic Release Inventory Chemicals

Not applicable.

##### Accident Precaution Chemicals

Not applicable.

##### Dangerous Substances Safety Management Act:

Not Applicable to Dangerous Materials

##### Waste Management Law

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Industrial waste

**The components of this product are reported in the following inventories:**

**United States TSCA Inventory :**

All substances listed as active on the TSCA inventory

**Canadian Domestic Substances List (DSL) :**

All components of this product are on the Canadian DSL.

**Australia. Australian Industrial Chemicals Introduction Scheme (AICIS) :**

On the inventory, or in compliance with the inventory

**New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :**

not determined

**Japan. ENCS - Existing and New Chemical Substances Inventory :**

On the inventory, or in compliance with the inventory

**Korea. Korean Existing Chemicals Inventory (KECI) :**

On the inventory, or in compliance with the inventory

**Philippines Inventory of Chemicals and Chemical Substances (PICCS) :**

On the inventory, or in compliance with the inventory

**China Inventory of Existing Chemical Substances :**

On the inventory, or in compliance with the inventory

**Taiwan Chemical Substance Inventory :**

On the inventory, or in compliance with the inventory

### 16. OTHER INFORMATION

Sources of key data used to compile the Safety Data Sheet  
Globally Harmonized System of Classification and Labelling of Chemicals (GHS)  
IARC: (International Agency for Research on Cancer)  
US. National Toxicology Program (NTP) Report on Carcinogens  
ECHA List of Publishable Substances Registered  
EU HPVCs (High Production Volume Chemicals)

|              |                      |
|--------------|----------------------|
| Issuing date | : 16.02.2021         |
| Version      | : 1.0                |
| Prepared by  | : Regulatory Affairs |

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.